Exercise 17: Synthesizing Balance Control

- You will use Synopsys Design Compiler to synthesize your **balance_cntrl** block, which has children PID & SegwayMath

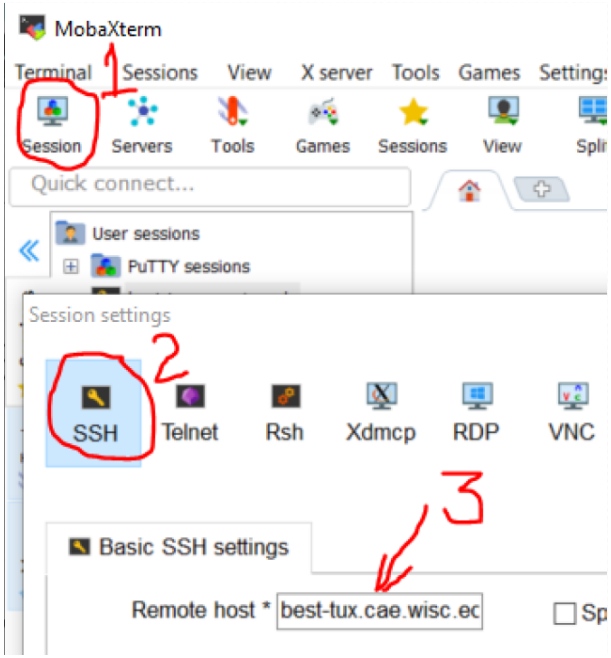
- Write a synthesis script (**balance_cntrl.dc**) that performs the following:
  - Defines a clock of 125MHz frequency and sources it to clock
  - Performs a set don't touch on the clock network
  - Defines input delays of 0.3 ns on all inputs other than clock
  - Defines a drive strength equivalent to a 2-input nand of size 2 from the Synopsys 32nm library (NAND2X2_RVT) for all inputs except clk and rst_n
  - Defines an output delay of 0.75ns on all outputs.
  - Defines a 50fF load on all outputs.
  - Sets a max transition time of 0.15ns on all nodes.
  - Employs the Synopsys 32nm wire load model for a block of size 16000 sq microns
  - Compiles, then flattens the design so it has no hierarchy, and compiles again.
  - Produces a min_delay report
  - Produces a max_delay report
  - Produces an area report
  - Writes out the gate level verilog netlist (**balance_cntrl.vg**)

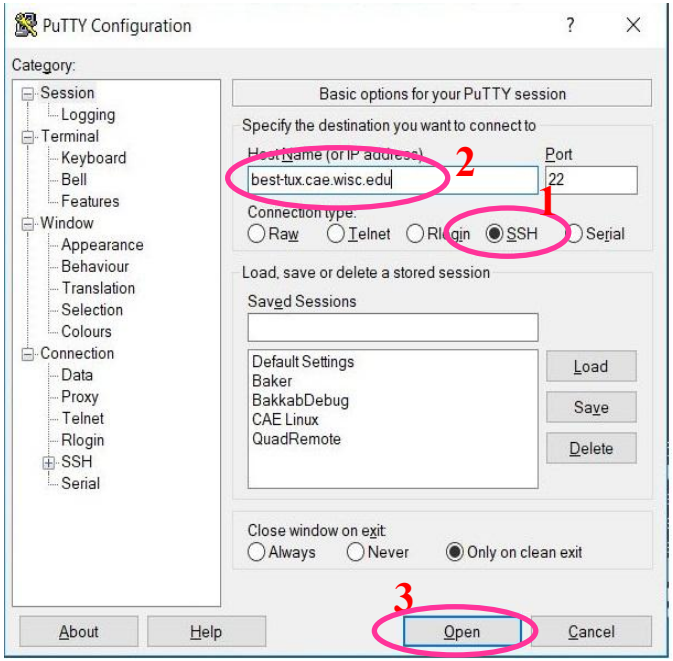
Note: Library default units are pF not fF

For reference my area was 3250ish square microns

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- You will use Synopsys Design Compiler to synthesize your **balance_cntrl** block, which has children PID & SegwayMath
- Design compiler is only on Linux machines, so you will need to remote into them for this exercise
  - Windows CAE machines have MobaXterm or PuTTy





Exercise 17: Synthesizing Balance Control

- Create a Ex17 folder for your synthesis

```
mkdir ex17
```

- Copy your balance_cntrl.sv, PID.sv, SegwayMath.sv into the folder
- cd to the folder and launch design compiler:

```
cd ex17
module load synopsys/suite
module load saed32_edk
synopsys-run
dc_shell
```

- Run your commands to synthesize balance_cntrl
  - It is recommended that you have a script to keep track of all the constraints, but run each line one at a time at first to make sure you catch any error messages

- **Submit:**
  - Your synthesis script (balance_cntrl.dc)
  - The output reports for area and timing (area.rpt, timing_min.rpt, timing_max.rpt)
  - The gatel level Verilog netlist (balance_cntrl.vg)